

Plagioclase An86 (MINTS habit): crystal morphology

Habit of the MINTS (Minerals In Thin Section, University of the Western Cape) plagioclase drawings, on an An86 cell

Plagioclase of composition An86 with the habit shown in the MINTS plagioclase drawings: a prism elongate along c , bounded by $\{010\}$, $\{110\}$ and $\{1\bar{1}0\}$ and terminated by sloping $\{001\}$ faces. Proportions were matched by eye to the bytownite drawing.

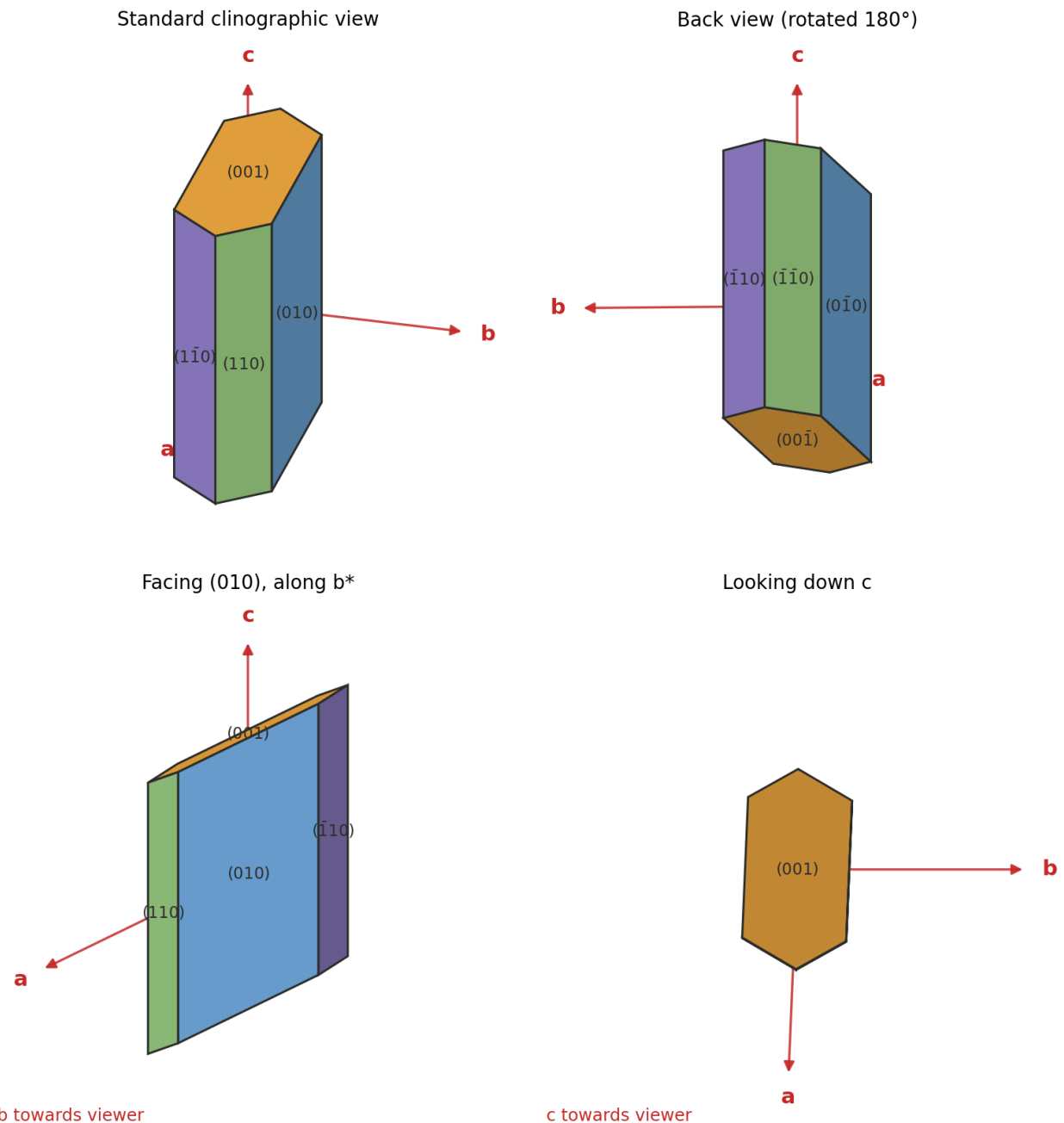


Figure 1. Orthographic views of the crystal in the frame used by SHAPE and the pygmsh classes (c along z , b in the yz plane, centre at the origin). Red arrows are the crystal axes; faces with the same colour belong to the same form.

Crystal data

Quantity	Value
SHAPE file	An86_MINTS_habit.SHP (title An86MINTS, class code 2B1)
Unit cell a, b, c (Å)	8.178, 12.87, 14.19
α , β , γ (°)	93.5, 115.9, 90.63
Point group / morphology symmetry	-1 (triclinic, centrosymmetric)
Faces, corners, edges	8, 12, 18
Angle between c and the (001) normal	26.21°
pygmsh class	plagioclase_An86_mints (xtal_geometries_feldspars.py)

Forms

Form	Same face, 7 Å axes	Central distance	Faces	Share of surface
{0 1 0}	{0 1 0}	0.42857	2	42.1%
{0 0 1}	{0 0 1}	1.00000	2	21.9%
{1 -1 0}	{1 -1 0}	0.71429	2	18.8%
{1 1 0}	{1 1 0}	0.71429	2	17.2%

Central distances are those in the SHAPE file (arbitrary units). Surface shares are computed from the polyhedron built from these forms, which reproduces SHAPE's own corners to 10^{-7} .

Size and shape when the pygmsh class is called with $z = 1$

Quantity	Value
Scaling	sf = z / fsp_z with fsp_z = 1.516215 (half-length along c in SHAPE units)
Length along c	2.000 (spans -1 to +1 along z)
Width (perpendicular to c, in (010))	1.085
Thickness (perpendicular to (010))	0.565
Length : width : thickness	3.54 : 1.92 : 1
Half-extents x , y , z	0.543, 0.298, 1.000
Volume V1	0.7680
Surface area	5.3216
Equivalent-sphere radius	0.5681
Surface / volume	6.930

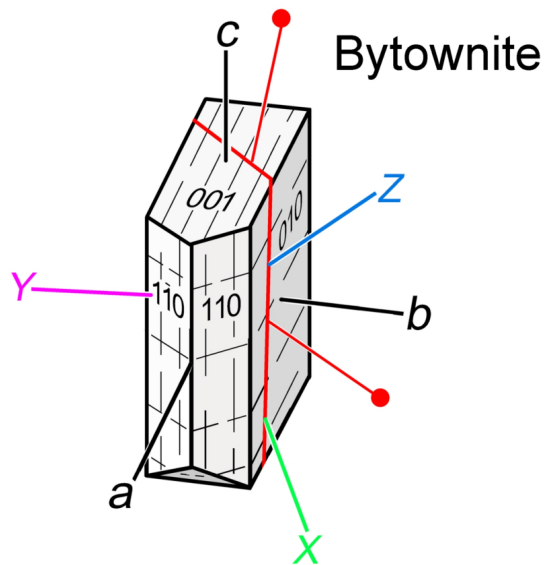
To match a measured length L along c use $z = L/2$; to match a measured volume V use $z = (V / V1)^{1/3}$.

Source and notes

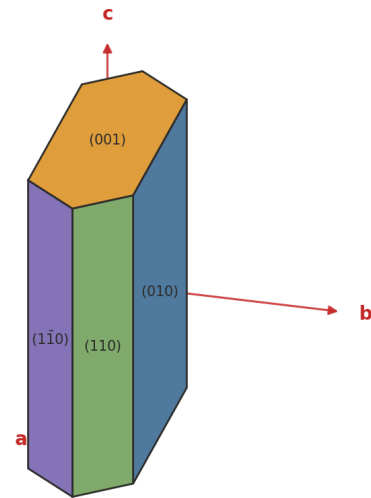
- Cell: bytownite (Wikipedia): a, b, c = 8.178, 12.870, 14.187 Å; α , β , γ = 93.5, 115.9, 90.63°, used for An86. For these four faces the symbols are the same on the 14 Å and 7 Å axes.
- Distances: scaled so that (001) = 1.000, as in the orthoclase SHAPE files 656 and 657. The proportions (length : width : thickness about 3.5 : 1.9 : 1) were chosen by eye; the drawings give no face distances.
- The same MINTS drawing is used for every composition from albite to anorthite; it illustrates optical orientation, not exact morphology.
- Scaling in the pygmsh class: fsp_z = 1.516215, the half-length along c.

Comparison with the source drawing

MINTS plagioclase drawing (bytownite)



Model, same viewing direction



Left: MINTS plagioclase drawing (bytownite). Right: the model rendered from the same standard viewing direction.

SHAPE input file (crystal definition)

```
An86MINTS
2B1      0      1  3.10      0
8.1780000 12.8700000 14.1870000
93.5000000 115.9000000 90.6300000
0      0      1  1.00000  0  0  0  0
0      1      0  .42857  0  0  0  0
1      1      0  .71429  0  0  0  0
1      -1     0  .71429  0  0  0  0
99      99     99  1.00000  0  0  0  0
```

Line 1: title. Line 2: crystal class code (2 = triclinic -1) and SHAPE options. Lines 3-4: a, b, c and alpha, beta, gamma. Following lines: h, k, l and central distance of each form, ending with 99 99 99. The display settings that follow in the file are not shown.